

Is inhibition involved in the processing of opaque compound words?

A study of individual differences

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We examined whether inhibition skills were recruited during the processing of compound words. Using an individual differences perspective, we analyzed whether participants' scores on the Stroop test predicted performance on lexical decision tasks involving compound words varying in their level of semantic opacity. The results show that inhibition is involved in the comprehension of fully opaque (e.g., *hogwash*) and fully transparent (e.g., *blueberry*) compound words, but we found no evidence for such an effect in the comprehension of partially opaque compound words (e.g., *strawberry*, *jailbird*).

Keywords: compound words, opaque, inhibition, individual differences, semantic transparency

In English, compound words (e.g., *blueberry*) consist of two unbound morphemes, the second being referred to as the head (e.g., *berry*) and the first being referred to as the modifier (e.g., *blue*). Compound words vary in semantic transparency (i.e., the degree to which the meanings of the constituents contribute to the meaning of the overall word). Depending on whether both, only one, or none of the constituents is semantically transparent, compound words can be classified into three categories: fully transparent (e.g., *blueberry*), partially transparent (e.g., *jailbird*, *strawberry*) or fully opaque (e.g., *hogwash*). This classification of the subtypes of compound words has led to the discovery of one of the most well-established findings in the compound word literature: namely, that opaque compound words are processed differently from transparent compounds. In a key study, Sandra (1990) demonstrated that transparent compounds were processed more quickly when they were preceded by a semantically related rather than by an unrelated

prime matched in length and frequency (e.g., people responded more quickly to *blueberry* if it was preceded by *red* versus *eye*). In contrast, opaque compounds did not benefit from the presence of a semantically related prime (e.g., people did not respond faster to *hogwash* if it was preceded by *pig* versus *lip*). Several subsequent studies have also shown that transparent compounds are responded to more quickly than opaque compounds (e.g., Ji, Gagné, & Spalding, 2011; Libben, Gibson, Yoon, & Sandra, 2003). This need for additional processing time has also been supported by eye-tracking data, which indicate that opaque compounds receive longer gazes than transparent compounds (Juhasz, 2007), as well as by typing data, which suggest that opaque constituents are typed more slowly on the computer keyboard than transparent constituents (Gagné & Spalding, 2014). A recent study using magnetoencephalography (Brooks & Cid del Garcia, 2015) indicated that the brain-response patterns for transparent and opaque compounds are different: Only transparent compounds show increased activity from 250 to 470 ms in the left anterior temporal lobe – a region that seems to be involved in morphological composition. Finally, encephalography data (MacGregor & Shtyrov, 2013) showed that only high-frequency opaque compounds are associated with larger MMN (Mismatch Negativity), which is interpreted as an indication that frequently used opaque compounds are accessed as a unitary representation from a lexical storage, rather than being computed via combinatorial mechanisms. Despite the abundance of studies showing the differences in processing times between transparent and opaque compound words, there is relatively little work that explores the cognitive mechanisms that cause this difference. Thus, the primary objective of this study is to address this matter that has largely been underexamined, and specifically investigate whether cognitive inhibition is a factor that could explain why opaque compounds need extra processing time.

As an initial attempt to explain the origin of the processing time difference between opaque and transparent compounds, Libben (2005) advanced the hypothesis that “the effects of semantic opacity, when they do occur, are related to the extra processing that is required to deactivate spuriously activated constituents” (p. 276). In other words, Libben suggests that opaque compound words require *cognitive inhibition* (i.e., the ability to suppress stimuli that are irrelevant to the task at hand), a capacity that ensures the efficient functioning of the working memory, which, otherwise, would be constantly overwhelmed by distracting information that would impede people from reaching their goals (Hasher & Zacks, 1988; Hasher, Zacks, & May, 1999). This cognitive process is essential for people’s good adjustment to everyday life; for instance, deficits in inhibition are associated with memory problems (Engle, Kane, & Tuholski, 1999), reckless driving (Sani, Tabibi, Fadardi, & Stavrinou, 2017), drug abuse (Monterosso, Aron, Cordova, Xu, & London, 2005), Attention Deficit Disorder (Bjorklund & Harnishfeger, 1990),

Obsessive-Compulsive Disorder (Enright & Beech, 1990), schizophrenia (Frith, 1979) and depression (Joormann, Yoon, & Zetsche, 2007).

In the field of psycholinguistics, inhibition has been shown to be involved in the comprehension of polysemous words (i.e., words that have more than one meaning) and figurative language, such as metaphors and idioms. In the case of sentences including polysemous words (e.g., *The dog gave a loud bark*), people need to inhibit the alternative context-inappropriate meanings of the ambiguous word (i.e., *layer of stems and roots of woody plants*) (Seidenberg, Tanenhaus, Leiman, & Bienkowski, 1982). Similarly, when people encounter a piece of figurative language such as *Lawyers are sharks*, they need to inhibit the irrelevant literal meaning *fish with cartilaginous skeleton* to compute the real intended meaning of the metaphor (Glucksberg, Newsome, & Goldvarg, 2001). When this inhibition fails as a consequence of disorders such as schizophrenia or Alzheimer's disease, people provide literal interpretations for figurative language (Iakimova, Passerieux, & Hardy-Bayle, 2006).

Given that, by definition, opaque compound words have constituents whose meanings are irrelevant to the computation of the whole word's meaning, Libben's (2005) assumption appears to be valid. It is logical to think that people have to inhibit the meanings of *hog* and *wash* to access the real meaning of *hogwash*. Furthermore, his claim is indirectly confirmed by the fact that when inhibition fails in patients suffering from aphasia, they tend to paraphrase opaque compound words as if they were transparent (e.g., *strawberry* might be incorrectly paraphrased as *a berry that has a straw*) (Libben, 1993). In support of Libben's inference, Ji et al. advanced the following explanation:

The absence of links between the opaque constituent and the compound (e.g., between *straw* and *berry*) is assumed to result in inhibition ..., and this inhibition ... prevents the meaning of the opaque constituent from interfering with the interpretation of the compound (e.g., the meaning of *straw* is inhibited). As a result, transparent compounds should receive more activation from their constituents (both at the lexical and semantic level) than would opaque compounds (2011, p. 412).

Given that Libben's (2005) claim has not yet been directly tested, there is to date a lack of direct evidence that could support his idea that, indeed, inhibition is necessary for comprehending opaque compound words. Thus, the goal of this study is to examine whether the language system needs to recruit inhibition when processing opaque compound words in order to deal with the interference originating from the irrelevant meanings of their constituents. To address this issue, we conducted two individual-difference experiments, in which we measured participants' scores on an inhibition task and analyzed whether these scores predicted their performance on lexical decision tasks that involved the comprehension of different types of compound words. The individual-difference methodology

allows us to analyze the entire range of the sample variability across the spectrum, which is often ignored by the flattening effects of working only with group means. It is also worth noting that this methodology, although seldom used in the compound word literature, has revealed previously ignored variables that influence the processing of compounds. For instance, Schmidtke, Van Dyke and Kuperman (2018) revealed that people's reading experience (i.e., amount of exposure to printed materials) and vocabulary size interact with semantic transparency to influence the pattern of eye movements when reading compound words. In addition, Kuperman and Van Dyke (2011) also showed that the profile of a person's verbal abilities (e.g., his or her ability to segment words and his or her level of reading comprehension) affects the recognition of a morphologically complex word (e.g., *trucker*) as a whole and via its decomposed morphemes (e.g., *truck* + *er*). It is worth pointing out, however, that these few individual-difference studies on compound words share one limitation: they are focused specifically on language-related skills and neglect more general cognitive abilities (e.g., memory or executive functions).

The current investigation consists of two experiments. In Experiment 1, participants performed an inhibition test and a lexical decision task with fully opaque (e.g., *hogwash*) and fully transparent (e.g., *cheekbone*) compound words. In Experiment 2, participants performed an inhibition test and a lexical decision task with partially opaque (e.g., *jailbird*, *strawberry*) compound words. We hypothesized that participants who experience difficulties (i.e., slow reaction times or low accuracy) when doing the inhibition task would also show difficulties specifically when understanding fully or partially opaque compound words, but not when understanding fully transparent compounds.

Experiment 1

The goal of Experiment 1 is to determine whether inhibition is involved in the comprehension of fully opaque, but not fully transparent compound words. Recall that the constituents of opaque compounds (e.g., *hum* and *bug*) do not contribute the meaning of the compound (e.g., *humbug*) and, thus, any meaning constructed for these constituents would be in conflict with the true meaning of the compound. Participants performed two tasks: a language task in which they had to comprehend fully opaque and fully transparent compounds, and an inhibition task in which they had to respond to visual stimuli, ignoring surrounding irrelevant information. If only opaque compounds require inhibition, participants who obtain low scores on the inhibition task will perform poorly only when responding to fully opaque compounds, but not when responding to fully transparent compounds.

Methods

Materials

Lexical decision task

The stimuli consisted of twenty-two fully transparent compound words (e.g., *cheekbone*), twenty-two fully opaque compound words (e.g., *hogwash*) and a total of one hundred and thirty-two fillers. We used three types of fillers. The first type of fillers consisted of forty-four monomorphemic words (e.g., *gratitude*) that were matched to the compound words in length and frequency¹ using SUBTLEX-US corpus (Brysbaert & New, 2009). The second type of fillers consisted of forty-four non-words that emulated a monomorphemic structure and that matched the compound words in length (e.g., *terisma*). The third type of fillers consisted of forty-four non-words that emulated a compound-like structure (i.e., they seemed to have two constituents) and that matched the compound words in length (e.g., *sidewolf*). To make sure that these eighty-eight non-word fillers were phonologically legal, they were created by combining syllables from words that appeared in English short stories. Each participant saw all the items, which are presented in Appendix A.

Inhibition task

A version of the Stroop test, which is a widely used measure of automatic inhibition (Stroop, 1935), was created in Supercard. The names of eleven colors (red, green, yellow, black, blue, orange, purple, pink, brown, gray and white) were displayed in ten different font colors (red, green, yellow, black, blue, orange, purple, pink, brown and gray). The test had a total of 200 trials: for half of them, the name of the color and its font matched (congruent items), and for the other half, the name of the color and its font did not match (incongruent items).

Procedure

All participants performed both the lexical decision task and the Stroop test. The order of the tasks was counterbalanced.

Lexical decision task

Participants indicated, as quickly as possible, whether an item was an English word or not. They pressed the key “j” if the item was a word, or the key “f” if it was a non-word. The order of the items was randomized for each participant. The

1. There were two words (*swansong* and *poppyseed*) whose frequency did not appear in SUBTLEX-US. We re-ran all the analyses presented in this article after removing these two items and we confirmed that none of results was impacted.

items were presented one at a time at the center of the screen. Each trial began with the message “Ready?” and the participants had to press the spacebar on the keyboard to see each item.

Inhibition task

Participants indicated, as quickly as possible, whether the name and font of the color were the same or not. They pressed the key “j” if they were the same, or the key “f” if they were different. This variation of the Stroop task has several advantages over the standard version: First, it is easier to administer because it does not require recording vocal responses; second, it is not affected by certain conscious strategies that participants who are familiar with the task may use to avoid interference (Bugg, Jacoby, & Toth, 2008; Logan, Zbrodoff, & Williamson, 1984), such as looking at the edge of each word to avoid reading it and to be able to focus solely on the color; and third, this variation yields the same type of interference as the standard Stroop task (Durgin, 2000; Treisman & Fearnley, 1969). The order of the items was randomized for each participant. The items were presented one at a time at the center of the screen. The experiment began with the message “Ready?” and the participants pressed the spacebar on the keyboard to see the first item.

Participants

Forty-one participants were recruited from Introductory Psychology classes in exchange for partial course credit. All participants were native speakers of English. Two participants were removed from all analyses: one due to low accuracy rate (less than 75%) on the lexical decision task and the other due to highly fluctuating reaction times (from 361 to 27,021 ms) on the Stroop test.

Results and discussion

Lexical decision task

Responses with reaction times that were more than 2.5 standard deviations greater than each participant’s mean were removed. Additionally, reaction times greater than 2500 ms were removed. The total number of observations removed represented 0.86% of the original data set. Inverse response time (i.e., $-1000/\text{RT}$) was used for the reaction time analysis because the Q-Q plots revealed that the inverse transformation was better at correcting for skewness in the residuals than was the log transformation. The response time analysis was based on correct trials only.

Separate analyses for the response time and accuracy data were conducted using linear mixed-effect (LME) analyses (Pinheiro & Bates, 2000) in Stata 13 using the function *mixed* for reaction time data and the function *meqrlogit* for the binary (correct vs. incorrect) accuracy data. Because these functions use dummy

coding for categorical variables, the coefficients (i.e., estimates) from these models correspond to simple effects, rather than to main effects. Therefore, the *contrast* function in Stata was used to test the main effects and interactions. The results of these tests are Wald tests and are reported as a chi-square. A Wald test is an approximation of a likelihood ratio test, but is more generalizable (i.e., it can be used in more situations than a likelihood ratio test) and indicates whether the factor (in the case of a main effect) or the interaction improves the fit of the model relative to a model that does not include that factor or interaction.

Type of compound (TT vs. OO), frequency of the compound and length of the compound word were entered as fixed effects and participant and item were included in the models as random effects. Neither the response time analysis nor the accuracy analysis showed an influence of compound type. Participants were as fast when responding to opaque-opaque compounds ($M=818$ ms, $SE=10$) as when responding to transparent-transparent compounds ($M=795$ ms, $SE=9$), $X^2(1)=0.71$, $p=.40$. Also, they were equally accurate when responding to opaque-opaque compounds ($M=87\%$, $SE=1.1$) as when responding to transparent-transparent compounds ($M=96\%$, $SE=0.6$), $X^2(1)=1.75$, $p=.19$.

Inhibition task

Responses with reaction times that were more than 2.5 standard deviations greater than each participant's mean (2.6% of the original data set) were removed. Inverse response time (i.e., $-1000/RT$) was used for the reaction time analysis because the Q-Q plots revealed that the inverse transformation was better at correcting for skewness in the residuals than was the log transformation. The response time analysis was based on correct trials only.

The data were analyzed using linear mixed effects regression models. Participant and item (i.e., color of the text) were included in the models as random effects, and congruency between the name of the color and its font as a fixed effect.

Surprisingly, participants were more accurate when responding to incongruent trials ($M=98\%$, $SE=2.5$) than responding to congruent trials ($M=95\%$, $SE=3.4$), $X^2(1)=32.69$, $p<.0001$. Participants were slower when responding to incongruent trials ($M=688$ ms, $SE=3.25$), compared to congruent trials ($M=651$ ms, $SE=2.90$), $X^2(1)=141.69$, $p<.0001$. These results suggest that participants might have engaged in a speed-accuracy trade-off, favouring accuracy over speed.

Lexical decision and inhibition task

For each participant, we computed the mean reaction time on congruent trials and on incongruent trials on the inhibition task. Then, for each participant we created two scores: an *inhibition score* and an *inhibition ratio*. The inhibition score was calculated by subtracting the mean reaction time on congruent trials

from the mean reaction time on incongruent trials. The inhibition score, also called Stroop interference, has traditionally been used in the literature (MacLeod, 1991) as a measure of the ability to suppress goal-irrelevant information. However, this measure has received criticism lately (Knight & Heinrich, 2017) because it may not account for generalized slowness and thus may not provide a pure interference measure. For instance, the absolute difference of 10 ms coming from a participant who took 110 ms on the incongruent trial and 100 ms on the congruent trial is more important than the same absolute difference between 2510 ms and 2500 ms coming from a participant who is generally slower. Therefore, in order to adjust the comparison and take into account the proportion of the difference, we also calculated the inhibition ratio by dividing each participant's mean reaction time on incongruent trials by the mean reaction time on congruent trials. The data were analyzed using LME analyses using the function *mixed* for reaction time data on Stata 13. The response time analysis was based on correct trials only.

Inhibition scores and reaction times on the lexical decision task

Participant and item were included in the models as random effects, and inhibition scores, type of compound (TT vs OO), frequency of the compound and length of the compound word as fixed effects. Figure 1 shows the distribution of the inhibition scores used to predict performance on the lexical decision task. The standardized regression coefficients, standard errors, tests of slopes, and confidence intervals for the model are shown in Table 1. The term "Constant" refers to the y-intercept.

Table 1. Standardized regression coefficients, standard errors, tests of slopes, and confidence intervals from the model using compound type, inhibition scores, frequency, and length to predict inverse-transformed reaction times from the lexical decision task in Experiment 1

	b	SE	z	p	95% confidence interval	
					Lower bound	Upper bound
Compound type	-0.025	0.038	-0.65	0.513	-0.098	0.049
Inhibition score	0.001	0.001	2.31	0.021	0.000	0.002
Compound type x Inhibition score	-0.000	0.000	-0.56	0.578	-0.001	.0004
Frequency	-0.019	0.007	-2.89	0.004	-0.032	-0.006
Length	-0.025	0.016	-1.59	0.112	-0.055	0.006
Constant	-1.150	0.138	-8.35	0.000	-1.420	-0.880

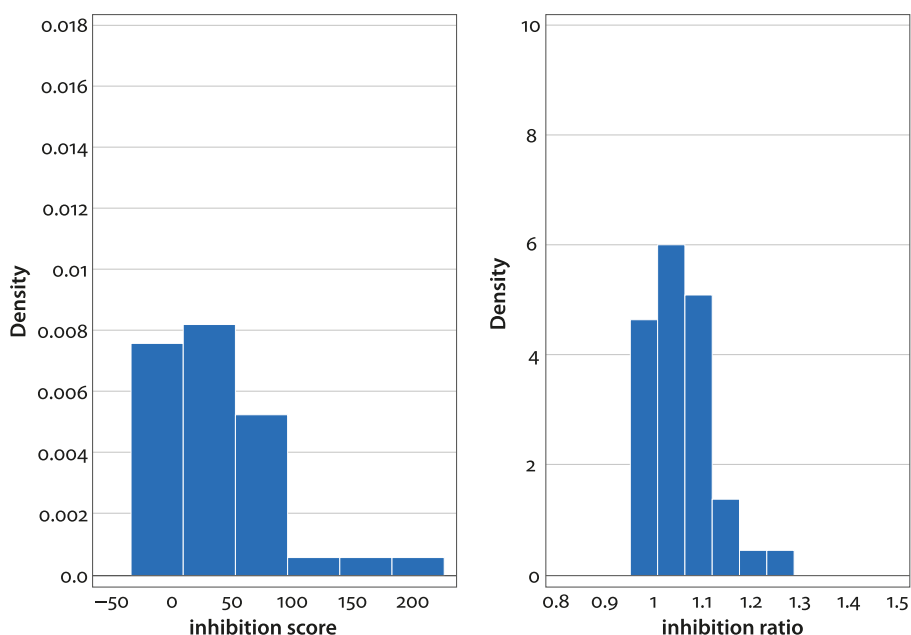


Figure 1. Histogram of inhibition scores and inhibition ratios used to predict performance on the lexical decision task of Experiment 1

Compound type and inhibition scores did not interact, $X^2(1) < 1$. Instead, there was a main effect of inhibition scores, $z = 2.25$, $p = .025$. Participants who had strong inhibition skills (i.e., participants who showed smaller differences between the time they took to respond to incongruent trials and congruent trials on the Stroop task) responded more quickly to both opaque and transparent compounds, compared to participants who had poor inhibition skills. In terms of the accuracy analysis, compound type and inhibition scores did not interact, $X^2(1) < 1$ and, unlike in the response time analysis, there was no influence of inhibition scores, $z = 0.90$, $p = .367$.

In summary, the lack of interaction between the type of compound word and the inhibition scores is contrary to our initial hypothesis that inhibition would be involved exclusively in the processing of opaque but not in the processing of transparent compound words. This matter will be further discussed in the General Discussion.

Inhibition ratio and reaction times on the lexical decision task

Participant and item were included in the models as random effects, and inhibition ratio, type of compound, frequency and length of the compound word as fixed effects. Figure 1 shows the distribution of the inhibition ratios used to predict

performance on the lexical decision task. The standardized regression coefficients, standard errors, tests of slopes, and confidence intervals for the model are shown in Table 2. Compound type and inhibition ratio did not interact, $X^2(1) < 1$. However, there was a main effect of inhibition ratio, $z = 2.02$, $p = .043$; participants who had strong inhibition skills (i.e., participants who showed smaller inhibition ratios on the Stroop task) responded more quickly compared to participants who had poor inhibition skills. In terms of the accuracy analysis, compound type and inhibition scores did not interact, $X^2(1) < 1$ and, unlike in the response time analysis, there was no influence of inhibition scores, $z = 0.57$, $p = .57$.

Table 2. Standardized regression coefficients, standard errors, tests of slopes, and confidence intervals from the model using compound type, inhibition ratio, frequency, and length to predict inverse-transformed reaction times from the lexical decision task in Experiment 1

	b	SE	z	p	95% confidence interval	
					Lower bound	Upper bound
Compound type	0.163	0.227	0.72	0.472	-0.282	0.609
Inhibition ratio	0.939	0.433	2.17	0.030	0.091	1.788
Compound type x Inhibition ratio	-0.184	0.212	-0.86	0.387	-0.600	0.233
Frequency	-0.019	0.007	-2.89	0.004	-0.032	-0.006
Length	-0.025	0.016	-1.59	0.112	-0.055	0.006
Constant	-2.094	0.478	-4.38	0.000	-3.030	-1.158

In summary, we found no evidence to suggest that inhibition is involved exclusively in the processing of opaque but not in the processing of transparent compound words. We will return to this issue in the General Discussion.

Experiment 2

Given that there is evidence that suggests that the processing of transparent-opaque and opaque-transparent compound words differs from the processing of fully opaque and fully transparent compounds (El-Bialy, Gagné, & Spalding, 2013), we investigated whether inhibition is involved in partially opaque compounds. In the current experiment, participants performed the same two tasks as in Experiment 1, but with partially opaque compound words.

Methods

Materials

Lexical decision task

The stimuli² consisted of twenty-two transparent-opaque compound words (e.g., *jailbird*), twenty-two opaque-transparent compound words (e.g., *strawberry*) and three types of fillers. The one hundred and thirty-two filler items were created following the same method used in Experiment 1. The items used are presented in Appendix B.

Inhibition task

The inhibition task was identical to the one used in Experiment 1.

Procedure

The procedures for both tasks were identical to the ones used in Experiment 1.

Participants

Forty-six participants, who had not participated in Experiment 1, were recruited from introductory psychology classes in exchange of partial course credit. All participants were native speakers of English. Three participants were removed from all analyses due to low accuracy rate (less than 75%) on the lexical decision task.

Results and discussion

Lexical decision task

Responses with reaction times that were more than 2.5 standard deviations greater than each participant's mean were removed. Additionally, reaction times greater than 2500 ms or smaller than 300 ms were also removed to eliminate the remaining outliers on the Q-Q plot. The total number of observations removed represented 1.26% of the original data set. Inverse response time (i.e., $-1000/\text{RT}$) was used for the reaction time analysis because the Q-Q plots revealed that the inverse transformation was better at correcting for skewness in the residuals than was the log transformation. The response time analysis was based on correct trials only.

2. There were five words (*catcall*, *bobbypin*, *pipedream*, *snailmail* and *knucklesandwich*) whose frequency did not appear in SUBTLEX-US. We re-ran all the analyses presented in this article after removing these five items and we confirmed that none of results was impacted.

The response time and accuracy data were analyzed separately using linear mixed-effect analyses. Participant and item were included in the models as random effects, and compound type, frequency of the compound, and length of the compound word were entered as fixed effects. Opaque-transparent compounds were more difficult to process than were the transparent-opaque compounds. The response time analysis indicated that participants were slower when responding to opaque-transparent compound words ($M=968$ ms, $SE=13$) than when responding to transparent-opaque compound words ($M=916$ ms, $SE=11$), $X^2(1)=6.52$, $p=.01$. The accuracy analysis revealed that participants were less accurate when responding to opaque-transparent compound words ($M=80\%$, $SE=1.3$) than when responding to transparent-opaque compound words ($M=88\%$, $SE=1.1$), $X^2(1)=7.33$, $p=.007$.

Inhibition task

Responses with reaction times that were more than 2.5 standard deviations greater than each participant's mean were removed. Additionally, reaction times greater than 2500 ms or smaller than 300 ms were also removed to eliminate the remaining outliers on the Q-Q plot. The total number of observations removed represented 2.86% of the original data set. Inverse response time (i.e., $-1000/RT$) was used for the reaction time analysis because the Q-Q plots revealed that the inverse transformation was better at correcting for skewness in the residuals than was the log transformation. The response time analysis was based on correct trials only.

The data were analyzed using LME analyses. Participant and item (i.e., color of the text) were included in the models as random effects, and congruency between the name of the color and its font as fixed effect. As in Experiment 1, participants were more accurate when responding to incongruent trials ($M=98\%$, $SE=2.3$) compared to congruent trials ($M=96\%$, $SE=3.0$), $X^2(1)=17.74$, $p<.0001$. Participants were significantly slower when responding to incongruent trials ($M=763$ ms, $SE=3.69$), compared to congruent trials ($M=703$ ms, $SE=3.58$), $X^2(1)=345.20$, $p<.0001$. These results suggest that congruency influenced processing but participants engaged in a speed-accuracy trade-off, as in Experiment 1.

Lexical decision and inhibition task

As in Experiment 1, we created an *inhibition score* and an *inhibition ratio* for each participant. The response time analysis was based on correct trials only. Figure 2 shows the distribution of the inhibition scores used to predict performance on the lexical decision task.

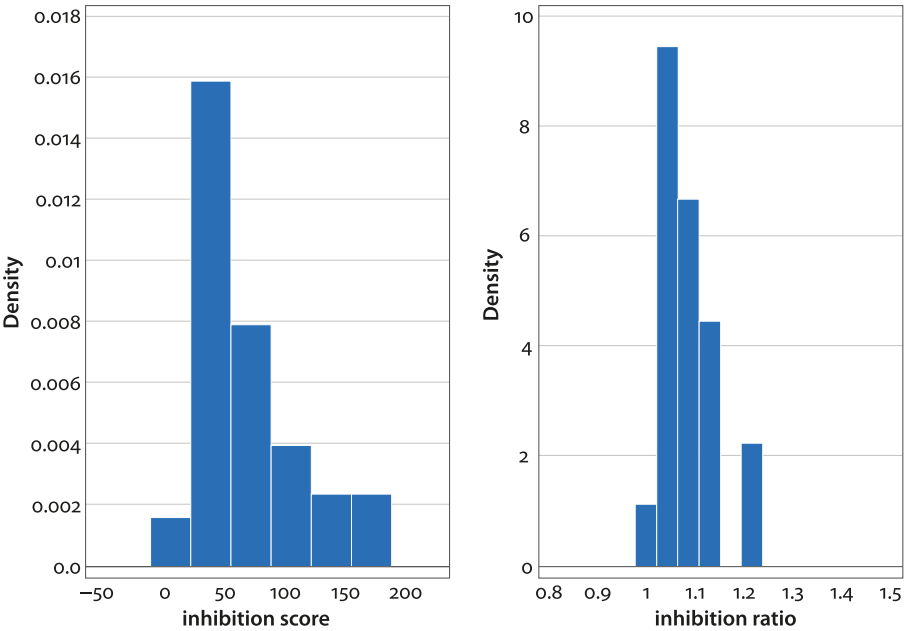


Figure 2. Histogram of inhibition scores and inhibition ratios used to predict performance on the lexical decision task of Experiment 2

Inhibition scores and reaction times on the lexical decision task

Separate analyses for the response time and accuracy data were conducted using linear mixed-effect (LME) analyses. Participant and item were included in the models as random effects, and inhibition scores, compound type, frequency and length were entered as fixed effects. The standardized regression coefficients, standard errors, tests of slopes, and confidence intervals for the model predicting response times are shown in Table 3. Compound type and the inhibition scores did not interact, $X^2(1) < 1$. Participants' inhibition skills were not related to their ability to process partially opaque compound words; inhibition score was not a valid predictor, $z = 0.22$, $p = .82$.³ In terms of the accuracy analysis, we also did not observe an interaction between compound type and inhibition score, $X^2(1) < 1$, nor did we observe an influence of inhibition ratio, $z = -1.21$, $p = .228$. Thus, there is no evidence that suggests that individual differences in inhibition skills influence the processing of partially opaque compound words.

3. We thank the reviewer who suggested that we should try fitting the model without the interaction term to verify whether the main effect of inhibition scores becomes significant. After removing the interaction term, the results did not changed: There was still no main effect of inhibition scores across all levels of type of compounds, $z = 0.23$, $p = .82$.

Table 3. Standardized regression coefficients, standard errors, tests of slopes, and confidence intervals from the model using compound type, inhibition scores, frequency, and length to predict inverse-transformed reaction times from the lexical decision task in Experiment 2

	b	SE	z	p	95% confidence interval	
					Lower bound	Upper bound
Compound type	-0.110	0.044	-2.53	0.012	-0.195	-0.025
Inhibition score	-0.000	0.001	-0.02	0.981	-0.001	0.001
Compound type x Inhibition score	0.000	0.000	0.74	0.459	-0.001	0.001
Frequency	-0.063	0.013	-4.89	0.000	-0.088	-0.037
Length	0.015	0.012	1.23	0.221	-0.009	0.039
Constant	-1.237	0.115	-10.80	0.000	-1.462	-1.013

Inhibition ratio and reaction times on the lexical decision task

Figure 2 shows the distribution of the inhibition ratios used to predict performance on the lexical decision task. Participant and item were included in the models as random effects, and inhibition ratio, type of compound word and frequency of length of the compound word as fixed effects. The standardized regression coefficients, standard errors, tests of slopes, and confidence intervals for the model predicting response times are shown in Table 4. We did not find any evidence to suggest that individual differences in inhibition skills influenced the processing of partially opaque compound words. Compound type and inhibition ratio did not interact, $X^2(1) = 1.41$, $p = .24$. Participants' inhibition skills were not related to their ability to process partially opaque compound words; inhibition ratio was not a successful predictor, $z = -0.62$, $p = .53$.⁴ In terms of the accuracy analysis, we also did not observe an interaction between compound type and inhibition ratio, $X^2(1) < 1$, nor did we observe an influence of inhibition ratio, $z = -1.40$, $p = .16$.

4. We thank the reviewer who suggested that we should try fitting the model without the interaction term to verify whether the main effect of inhibition ratio becomes significant. After removing the interaction term, our results have not changed: There was still no main effect of inhibition ratio across all levels of type of compounds, $z = -0.61$, $p = .54$

Table 4. Standardized regression coefficients, standard errors, tests of slopes, and confidence intervals from the model using compound type, inhibition ratio, frequency, and length to predict inverse-transformed reaction times from the lexical decision task in Experiment 2

	b	SE	z	p	95% confidence interval	
					Lower bound	Upper bound
Compound type	−0.464	0.316	−1.47	0.142	−1.083	0.155
Inhibition ratio	−0.491	0.507	−0.97	0.333	−1.485	0.503
Compound type x Inhibition ratio	0.343	0.289	1.19	0.236	−0.224	0.910
Frequency	−0.062	0.013	−4.89	0.000	−0.087	−0.037
Length	0.015	0.012	1.23	0.220	−0.009	0.039
Constant	−0.705	0.561	−1.26	0.209	−1.804	0.394

Comparing Experiment 1 and Experiment 2

The finding that in Experiment 1 the processing of fully opaque and fully transparent compounds was predicted by the performance on the inhibition task, whereas in Experiment 2 the processing of partially opaque compounds was not predicted by the performance on the inhibition task motivated us to further investigate the underlying causes of this difference. One possible explanation for these very different patterns of results is that participants performed more poorly on the lexical decision task of Experiment 2, compared to Experiment 1, which created a floor effect, leaving a smaller margin for the inhibition scores or rates to predict the performance on the lexical decision task.

To investigate whether partially opaque compound words (Experiment 1) are indeed more difficult to process than fully opaque or fully transparent compounds (Experiment 2), we combined the data sets from Experiment 1 and Experiment 2 and fit a linear mixed effects regression model using experiment, frequency and length of the compound word as fixed effects and participant and item as random effects. The reaction time analysis showed that participants were slower when responding to the partially opaque compound words from Experiment 2 ($M = 942$ ms, $SE = 8$), than to the fully opaque and the fully transparent compounds from Experiment 1 ($M = 806$ ms, $SE = 7$), $X^2(1) = 7.96$, $p = .005$. In terms of the control variables, word frequency influenced response time, $z = -4.03$, $p < .0001$, but length did not, $z < 1$. The accuracy analysis showed that participants were significantly less accurate when responding to the partially opaque compound words

from Experiment 2 ($M=84\%$, $SE=0.87$), than the fully opaque and the fully transparent compounds from Experiment 1 ($M=91\%$, $SE=.69$), $X^2(1)=5.46$, $p=.02$. Word frequency influenced accuracy, $z=2.27$, $p<.02$, but length did not, $z<1$. Thus, both the response time and accuracy analyses indicated that partially opaque compounds were more difficult to process than compounds for which the transparency of the two constituents were consistent (i.e., fully opaque and fully transparency compounds).

To further explore the role of constituent transparency, we fit a model with the semantic transparency of the first constituent, semantic transparency of the second constituent, interaction of the semantic transparency of the first constituent and the semantic transparency of the second constituent, word frequency of the compound and length of the compound word as fixed effects. Table 5 shows the standardized regression coefficients, standard errors, tests of slopes, and confidence intervals model for the model predicting response times and Table 6 shows the same information but for the model predicting the accuracy data.

Table 5. Standardized regression coefficients, standard errors, tests of slopes, and confidence intervals effects from the model using semantic transparency of the first constituent (semtransC1), semantic transparency of the second constituent (semtransC2), frequency, and length to predict inverse-transformed reaction times from the lexical decision task in Experiments 1 and 2 combined

	b	SE	z	p	95% confidence interval	
					Lower bound	Upper bound
SemtransC1	0.199	0.055	3.64	0.000	0.092	0.306
SemtransC2	0.131	0.053	2.48	0.013	0.027	0.234
SemtransC1 x SemtransC2	-0.270	0.094	-2.88	0.004	-0.454	-0.087
Frequency	-0.028	0.006	-4.52	0.000	-0.040	-0.016
Length	-0.001	0.010	-0.15	0.885	-0.022	0.019
Constant	-1.336	0.100	-13.30	0.000	-1.533	-1.139

For the response time analysis, the impact of a constituent's semantic transparency was influenced by the semantic transparency of the other constituent; semantic transparency of the first constituent and the semantic transparency of the second constituent interacted, $X^2(1)=8.32$, $p=.004$. We followed up on this interaction by examining the simple effect of the first constituent's transparency at each level of the second constituent's transparency. We observed an influence of the first constituent's semantic transparency only when the second constituent was transparent. When the second constituent was transparent, responses were slower when

Table 6. Standardized regression coefficients, standard errors, tests of slopes, and confidence intervals from the model using semantic transparency of the first constituent (semtransC1), semantic transparency of the second constituent (semtransC2), frequency, and length to predict accuracy (correct vs. incorrect) from the lexical decision task in Experiments 1 and 2 combined

	b	SE	z	p	95% confidence interval	
					Lower bound	Upper bound
SemtransC1	−2.187	0.592	−3.70	0.000	−3.347	−1.028
SemtransC2	−1.141	0.568	−2.01	0.044	−2.254	−0.028
SemtransC1 x SemtransC2	2.109	0.859	2.46	0.014	0.426	3.792
Frequency	0.301	0.106	2.84	0.005	0.093	0.509
Length	−0.037	0.146	−0.26	0.798	−0.323	0.249
Constant	4.209	1.395	3.02	0.003	1.476	6.943

the first constituent was opaque (OT) than when the first constituent was transparent (TT), $X^2(1) = 13.23$, $p < .001$. However, when the second constituent was opaque, response times for compounds with transparent first constituents (TO) did not differ from compounds with opaque first constituents (OO), $X^2(1) = 1.77$, $p < .18$.

An alternative way to break-down the interaction is to examine the simple effect of the second constituent’s transparency at each level of the first constituent’s transparency. The influence of the second constituent’s transparency depended on the semantic transparency of the first constituent. When the first constituent was transparent, responses were slower when the second constituent was opaque (TO) than when the second constituent was transparent (TT), $X^2(1) = 6.14$, $p < .013$. However, when the first constituent was opaque, responses were faster when the second constituent was opaque (OO) than when the second constituent was transparent (OT), $X^2(1) = 6.58$, $p < .010$.

In terms of the analysis of the accuracy data, we also found a significant interaction between the semantic transparency of the first constituent of the compound and the semantic transparency of the second constituent of the compound, $X^2(1) = 6.03$, $p = .01$. The analysis of the simple effect of the first constituent’s transparency at each level of the second constituent’s transparency, revealed that when the second constituent was transparent, responses were less accurate when the first constituent was opaque (OT) than when the first constituent was transparent (TT), $X^2(1) = 13.67$, $p < .001$. However, when the second constituent was opaque, accuracy did not differ for compounds with opaque first constituents (OO) and compounds with transparent first constituents (TO), $X^2(1) < 1$. In terms of the analysis of the simple effect of the second constituent’s transparency at each level of the first

constituent's transparency, when the first constituent was transparent, responses were less accurate when the first constituent was opaque (TO) than when the first constituent was transparent (TT), $X^2(1) = 4.04$, $p < .04$. When the first constituent was opaque, accuracy did not differ for compounds with opaque first constituents (OO) and compounds with transparent first constituents (OT), $X^2(1) < 2.86$, $p = .09$.

In summary, our first set of models indicated that partially opaque compound words are more difficult to process than fully transparent or fully opaque compounds. Therefore, the absence of an association between inhibition skills and the processing of partially opaque compounds reflected in Experiment 2 could be interpreted as a consequence of the additional processing challenges that opaque-transparent and transparent-opaque compounds pose. In fact, it is possible that this added processing difficulty overrides the influence of inhibition. However, even though these results are informative, they should certainly be taken with caution because, given that we tested two different groups of participants across our two experiments, it is possible that participants of Experiment 2 performed inherently worse than participants of Experiment 1, instead of partially opaque being more difficult to process than fully opaque and fully transparent compound words. For instance, the two groups of participants could significantly differ in terms of reading ability, motivation or other characteristics that could affect their performance (although it would be unusual for samples of participants this large to differ markedly enough in their characteristics to account for the magnitude of differences that we observed between Experiment 1 and Experiment 2). Moreover, it is also possible that differing sets of items across experiments could be differentially easy or hard to discriminate from non-words, which could lead to the participants setting different response thresholds.

Our second set of analyses provided a more fine-grained analysis of the role of constituent transparency on processing and also revealed that partially opaque compounds were more difficult to process. In particular, responses to opaque-transparent compounds were slower than responses to either transparent-transparent or opaque-opaque compounds and responses to transparent-opaque compounds were slower than responses to transparent-transparent compounds, but did not differ from opaque-opaque compounds. In addition, these analyses revealed both the response time and accuracy data indicated that the semantic transparency of the first constituent influenced ease of processing only when the second constituent was semantically transparent; that is, opaque-transparent compounds were more difficult to process than transparent-transparent compounds, but opaque-transparent compounds were not more difficult than opaque-opaque compounds.

General discussion

The results of Experiment 1 suggest that inhibition is involved in the comprehension of not only fully opaque, but also fully transparent compound words. Experiment 2, on the other hand, suggests that partially opaque compound words are inherently more difficult to process, compared to fully opaque and fully transparent compounds, and that this additional difficulty may override the influence of inhibition scores, which then stop being good predictors of the lexical decisions.

The association between inhibition skills and the processing of both fully opaque and fully transparent compound words found in Experiment 1 may have two different explanations. On one hand, as Libben (2005) stated, processing fully opaque compounds require inhibition in order to suppress the irrelevant meanings of both constituents, which interferes with the computation of the overall meaning. On the other hand, processing fully transparent compounds may require inhibition not because the meanings of their constituents are semantically opaque and need to be suppressed, but rather because fully transparent compounds are similar to polysemous words in the sense that readers need to decide which of the different possible meanings needs to be selected. For instance, in the same way that readers have to decide whether the polysemous word *bark* means *layer of stems and roots of woody plant* or *sharp sound a dog makes*, they have to choose the right relation that connects the two constituents of a fully transparent compound word (e.g., *snowball* as *ball MADE OF snow*, as *ball that IS snow* or as *ball that HAS snow*) and suppress the other possible relations. This hypothesis agrees with an idea that had already been suggested previously, namely that the comprehension of transparent compound words involves meaning computation and selection of relations (Gagné & Spalding, 2009). It is worth pointing out, however, that future research needs to determine whether inhibition per se modulates compound word processing or whether other verbal and non-verbal cognitive abilities that may covary with inhibition, such as general intelligence (Imbrosciano & Berlach, 2006), creativity (Benedek, Franz, Heene, & Neubauer, 2012) or spatial and verbal working memory (Traverso, Mantini, Usai, & Viterbori, 2015) are involved as well.

Findings from Experiment 2 indicate that there is not enough evidence to state that individual differences in inhibition skills are good predictors of the processing of partially opaque compound words. The analysis that compared Experiment 1 and Experiment 2 suggested that partially opaque compounds are inherently more difficult to process than fully transparent and fully opaque compounds. Therefore, it is possible that the lexical decision task with partially opaque compounds was too difficult to let the inhibition effect show. The idea that partially opaque compounds

are more difficult to process than fully opaque compounds may seem counterintuitive: The question that arises is why having one constituent that is transparent does not help the comprehension of the whole word. However, it is worth noting that previous research has already suggested that partially transparent compound words present a pattern of results that are different from fully transparent compounds. For instance, El-Bialy et al. (2013) have shown that the ease of processing a compound word depends not only on the opacity of the constituents, but also on whether the constituents have similar levels of semantic transparency. These authors found that a semantically related prime facilitated the processing of a compound when the transparency of the first and second constituents matched (i.e., when both were transparent or when both were opaque), but not when the transparency of the constituents differed, as in the case of partially opaque compounds (e.g., presenting *ear* helped the processing of *eyesight*, but not *eyetooth*). The authors explained these puzzling results as follows: When a semantically related prime is presented, a lexical activation of the constituents of the compound word occurs, and the language system attempts to construct a meaning; if the compound is fully transparent, the constructed meaning derived from the constituents is congruent with the real meaning, so it benefits from the presence of the prime; if the compound is fully opaque, there is a conflict between the meaning derived from the constituents and the real meaning of the compound, but given that these two meanings are so unrelated to each other the conflict is resolved easily and the benefit from the prime is not offset; however, if the compound is partially transparent, the meaning derived from the constituents partially overlaps with the real meaning of the whole word, which creates a more difficult conflict that requires additional time to be resolved, which results in the advantage from the prime being offset. Moreover, the fact that partially opaque behave differently than fully opaque or fully transparent compounds has been noted not only during comprehension tasks, but also during written production tasks. For instance, Gagné and Spalding (2016) showed that it is more difficult to start typing the second constituent of fully transparent compounds (e.g., *eggbeater*) than opaque-transparent compounds (e.g., *eggplant*). The authors explained this phenomenon as follows: When typing the second constituent of a compound word, participants' language system has to shift the attention from the first constituent and suppress it; when the compound is fully transparent, the meaning of the first constituent, which is related to the meaning of the whole word, is still strongly available when starting to type the second constituent, and this delays the typing process. Gagné and Spalding (2014) also showed that participants took longer to initiate the typing process when the compound was opaque-transparent, compared to when it was fully transparent, and when the compound was fully opaque, compared to when it was transparent-opaque. In other words, our study has confirmed the idea revealed by previous

research using partially opaque compounds that the language system is sensitive not only to the level of transparency of each of the constituents of a compound word, but also to whether the first and the second constituent have matching levels of semantic transparency. To summarize, we suggest that the semantic distance between the stored representation of a compound and the meanings that might be initially computed from the constituents affects the ease with which a person can settle on the actual meaning of the whole word. Thus, the computed meanings of fully opaque compounds might be so semantically distant from the stored meaning that there is relatively little competition, while the computed meanings of partially opaque (i.e., opaque-transparent and transparent-opaque) compounds will have some overlap because of the modifier and head respectively, which might make them semantically close enough to interfere with the stored meaning.

In both experiments, congruent trials are faster but less accurate than incongruent trials for the inhibition task. We suggest that this reflects a speed-accuracy trade-off. However, an alternative explanation might be due to hand dominance; the left hand is slower than the right hand, which should cause left-handed responses (i.e., incongruent trials) to be slower but more accurate than right-handed responses (i.e., congruent trials) because the participant has more time to correct his/her left-handed response. According to this explanation, hand dominance might explain why we found that there are more errors in the congruent condition (right-handed responses) than in incongruent conditions (left-handed responses). However, there are several reasons why hand dominance might not be a good explanation for the fact that congruent trials are faster but less accurate than incongruent trials. First, the congruency effect in tasks such as the Stroop task or the Simon task, is well-established and tends to be large (our congruency effect is approximately 60 ms). Handedness effects are generally small (approximately 10 ms or less) in simple tasks involving reaction time, and often not significant in more complex tasks; sometimes, reaction times can even be faster for the non-dominant hand (Bisiach, Mini, Sterzi, & Vallar, 1982; Gignac & Vernon, 2004). Furthermore, if the differences we observed in our experiments were due to hand dominance, the errors made by the faster right hand should correspond to the incongruent condition, not the congruent condition. This would occur because we must take into account that if an error is made in the congruent condition, the hand that responds is not the right hand, but the left (slower) one.

One additional issue is that Stroop tasks are sometimes taken to involve primarily conscious attempts at inhibition of the incorrect response, whereas the inhibition that could be involved in dealing with potential conflict between or among compound word meanings and the constituents would be primarily unconscious. This is particularly true of the standard Stroop paradigm, where one can “feel” oneself attempting to stop the incorrect response, and indeed in some

sense to stop reading the word. This contrasts strongly with the intuitive feel of competition among compound meanings and constituents, which seems more or less non-existent. However, there is also some reason to believe that even the standard Stroop paradigm might involve both conscious and unconscious inhibition, and the variation of the Stroop paradigm that we adopted, in which one is directly asked to say whether the word form and the color match or mismatch, avoids the more conscious aspects of the standard Stroop paradigm. In particular, one is not presented with the response conflict, and one cannot do the task by attempting not to read the word. In addition, much previous research has suggested the idea that there may not be a clear-cut distinction between “conscious” (controlled, voluntary or willful) and “unconscious” inhibition. For instance, in a study using fMRI, Horga and Maia (2012) have suggested that unconscious and conscious processes share many characteristics and, in particular, that even processes such as subliminal priming can affect cognitive control mechanisms. Nevertheless, it is possible that the variant of the Stroop task that we have chosen still depends to some degree on conscious inhibition, and this would tend to suggest that our data might be underestimating the effect of individual differences in inhibition skill on compound word processing.

In conclusion, we can state that the field that relates specific executive functions to the comprehension of different types of compound words remains relatively unexplored. However, this study has contributed to an initial understanding of this matter by pointing out two facts: on one hand, that inhibition skills may be a key factor in the processing of this type of complex words, and, on the other hand, that the reasons why the language system needs to recruit inhibition may depend on the level of semantic transparency of the compound: Fully opaque compound words seem to involve inhibition in order to suppress the semantically irrelevant meaning of the constituents, whereas fully transparent compounds seem to involve inhibition in order to decide which relation that connects the two constituents is the best.

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Appendix A

Stimuli presented in the lexical decision task in Experiment 1

String of letters presented	Type of stimulus
bandwagon	Fully opaque compound
blackmail	Fully opaque compound
blockbuster	Fully opaque compound
bootleg	Fully opaque compound
busybody	Fully opaque compound

String of letters presented	Type of stimulus
buttercup	Fully opaque compound
dingbat	Fully opaque compound
dumbbell	Fully opaque compound
fanfare	Fully opaque compound
hedgehog	Fully opaque compound
highlight	Fully opaque compound
hogwash	Fully opaque compound
honeymoon	Fully opaque compound
jackpot	Fully opaque compound
marshmallow	Fully opaque compound
mushroom	Fully opaque compound
nutcase	Fully opaque compound
scapegoat	Fully opaque compound
snapdragon	Fully opaque compound
stalemate	Fully opaque compound
swansong	Fully opaque compound
tightwad	Fully opaque compound
cheekbone	Fully transparent compound
clamshell	Fully transparent compound
cloverleaf	Fully transparent compound
corkscrew	Fully transparent compound
fingertip	Fully transparent compound
gunpowder	Fully transparent compound
hailstorm	Fully transparent compound
hairnet	Fully transparent compound
lampshade	Fully transparent compound
mincemeat	Fully transparent compound
molehill	Fully transparent compound
nosebleed	Fully transparent compound
paintbrush	Fully transparent compound
poppyseed	Fully transparent compound

String of letters presented	Type of stimulus
racehorse	Fully transparent compound
rosebud	Fully transparent compound
snakeskin	Fully transparent compound
thumbnail	Fully transparent compound
toothpick	Fully transparent compound
wastebasket	Fully transparent compound
windowpane	Fully transparent compound
wristwatch	Fully transparent compound
dominican	Non compound word filler
gratitude	Non compound word filler
realization	Non compound word filler
monarch	Non compound word filler
glaucoma	Non compound word filler
franchise	Non compound word filler
anchovy	Non compound word filler
literacy	Non compound word filler
nirvana	Non compound word filler
advocacy	Non compound word filler
vandalism	Non compound word filler
lactose	Non compound word filler
procedure	Non compound word filler
amnesia	Non compound word filler
electrician	Non compound word filler
diplomat	Non compound word filler
prodigy	Non compound word filler
diversity	Non compound word filler
conformism	Non compound word filler
actuality	Non compound word filler
tubercle	Non compound word filler
allegory	Non compound word filler
alchemist	Non compound word filler

String of letters presented	Type of stimulus
trapezoid	Non compound word filler
allocation	Non compound word filler
retrieval	Non compound word filler
emergence	Non compound word filler
itinerary	Non compound word filler
symbiosis	Non compound word filler
hipster	Non compound word filler
blaspheme	Non compound word filler
phosphate	Non compound word filler
affluent	Non compound word filler
meteorite	Non compound word filler
memorandum	Non compound word filler
mysticism	Non compound word filler
recipient	Non compound word filler
reptile	Non compound word filler
symposium	Non compound word filler
monograph	Non compound word filler
detergent	Non compound word filler
testimonial	Non compound word filler
recompense	Non compound word filler
attachment	Non compound word filler
ansition	Nonword filler with monomorphemic-like structure
vangeliss	Nonword filler with monomorphemic-like structure
untertracts	Nonword filler with monomorphemic-like structure
trovers	Nonword filler with monomorphemic-like structure
tronomic	Nonword filler with monomorphemic-like structure
ricultrur	Nonword filler with monomorphemic-like structure
megatlo	Nonword filler with monomorphemic-like structure
ervation	Nonword filler with monomorphemic-like structure
cologic	Nonword filler with monomorphemic-like structure
sourcest	Nonword filler with monomorphemic-like structure

String of letters presented	Type of stimulus
gisticoat	Nonword filler with monomorphemic-like structure
diamint	Nonword filler with monomorphemic-like structure
ervolemia	Nonword filler with monomorphemic-like structure
terisma	Nonword filler with monomorphemic-like structure
quenciality	Nonword filler with monomorphemic-like structure
trophims	Nonword filler with monomorphemic-like structure
assiert	Nonword filler with monomorphemic-like structure
milatiocy	Nonword filler with monomorphemic-like structure
complishly	Nonword filler with monomorphemic-like structure
phenylmon	Nonword filler with monomorphemic-like structure
dosteron	Nonword filler with monomorphemic-like structure
descenss	Nonword filler with monomorphemic-like structure
framplis	Nonword filler with monomorphemic-like structure
pressdite	Nonword filler with monomorphemic-like structure
tantaneous	Nonword filler with monomorphemic-like structure
tommutals	Nonword filler with monomorphemic-like structure
texteenex	Nonword filler with monomorphemic-like structure
niversiof	Nonword filler with monomorphemic-like structure
elfilment	Nonword filler with monomorphemic-like structure
plicits	Nonword filler with monomorphemic-like structure
mollgrass	Nonword filler with monomorphemic-like structure
tatchwors	Nonword filler with monomorphemic-like structure
effinoil	Nonword filler with monomorphemic-like structure
earthfulls	Nonword filler with monomorphemic-like structure
erfeinting	Nonword filler with monomorphemic-like structure
tractivis	Nonword filler with monomorphemic-like structure
tionwider	Nonword filler with monomorphemic-like structure
precial	Nonword filler with monomorphemic-like structure
ronitally	Nonword filler with monomorphemic-like structure
quelfiort	Nonword filler with monomorphemic-like structure
wrismatic	Nonword filler with monomorphemic-like structure

String of letters presented	Type of stimulus
ellitatus	Nonword filler with monomorphemic-like structure
letrerolit	Nonword filler with monomorphemic-like structure
mertractor	Nonword filler with monomorphemic-like structure
ragefiber	Nonword filler with compound-like structure
wittymain	Nonword filler with compound-like structure
womenswitch	Nonword filler with compound-like structure
postsoy	Nonword filler with compound-like structure
sidewolf	Nonword filler with compound-like structure
peopleraw	Nonword filler with compound-like structure
roadbud	Nonword filler with compound-like structure
vibewear	Nonword filler with compound-like structure
wigmale	Nonword filler with compound-like structure
agonygem	Nonword filler with compound-like structure
openlarva	Nonword filler with compound-like structure
vanring	Nonword filler with compound-like structure
flirtpoor	Nonword filler with compound-like structure
clogpaw	Nonword filler with compound-like structure
crestarctic	Nonword filler with compound-like structure
geekdawn	Nonword filler with compound-like structure
cowfrog	Nonword filler with compound-like structure
tractwrap	Nonword filler with compound-like structure
mildcounty	Nonword filler with compound-like structure
civilfull	Nonword filler with compound-like structure
lacegolf	Nonword filler with compound-like structure
drillfox	Nonword filler with compound-like structure
juiceloan	Nonword filler with compound-like structure
jeanmedal	Nonword filler with compound-like structure
devourheck	Nonword filler with compound-like structure
beanmoral	Nonword filler with compound-like structure
gospelear	Nonword filler with compound-like structure
anytechno	Nonword filler with compound-like structure

String of letters presented	Type of stimulus
ponypinky	Nonword filler with compound-like structure
limebeg	Nonword filler with compound-like structure
ovenlimbs	Nonword filler with compound-like structure
porchfolk	Nonword filler with compound-like structure
bondjude	Nonword filler with compound-like structure
clawalign	Nonword filler with compound-like structure
chartsense	Nonword filler with compound-like structure
dozerjust	Nonword filler with compound-like structure
hempopera	Nonword filler with compound-like structure
diveran	Nonword filler with compound-like structure
indexhonk	Nonword filler with compound-like structure
gloveyawn	Nonword filler with compound-like structure
smileeuro	Nonword filler with compound-like structure
worstsilica	Nonword filler with compound-like structure
gossipmaze	Nonword filler with compound-like structure
igloosuave	Nonword filler with compound-like structure

Appendix B

Stimuli presented in the lexical decision task in Experiment 2

String of letters presented	Type of stimulus
mayfly	Opaque-transparent compound
bobtail	Opaque-transparent compound
jaywalk	Opaque-transparent compound
beeline	Opaque-transparent compound
warthog	Opaque-transparent compound
crowbar	Opaque-transparent compound
catcall	Opaque-transparent compound
clubfoot	Opaque-transparent compound
joystick	Opaque-transparent compound
wormwood	Opaque-transparent compound

String of letters presented	Type of stimulus
alderman	Opaque-transparent compound
smallpox	Opaque-transparent compound
bobbypin	Opaque-transparent compound
crabgrass	Opaque-transparent compound
flagstone	Opaque-transparent compound
horseplay	Opaque-transparent compound
buckwheat	Opaque-transparent compound
butterfly	Opaque-transparent compound
pipedream	Opaque-transparent compound
snailmail	Opaque-transparent compound
strawberry	Opaque-transparent compound
frankincense	Opaque-transparent compound
hotdog	Transparent-opaque compound
inkwell	Transparent-opaque compound
oddball	Transparent-opaque compound
boldface	Transparent-opaque compound
sideburn	Transparent-opaque compound
typeface	Transparent-opaque compound
slowpoke	Transparent-opaque compound
jailbird	Transparent-opaque compound
sidekick	Transparent-opaque compound
doughnut	Transparent-opaque compound
blackhead	Transparent-opaque compound
lazybones	Transparent-opaque compound
turntable	Transparent-opaque compound
honeycomb	Transparent-opaque compound
gingersnap	Transparent-opaque compound
spoilsport	Transparent-opaque compound
chatterbox	Transparent-opaque compound
cheapskate	Transparent-opaque compound
lumberjack	Transparent-opaque compound

String of letters presented	Type of stimulus
doubleheader	Transparent-opaque compound
butterscotch	Transparent-opaque compound
knucklesandwich	Transparent-opaque compound
marmot	Non compound word filler
opiate	Non compound word filler
syrians	Non compound word filler
slavish	Non compound word filler
chicory	Non compound word filler
migrant	Non compound word filler
adrenal	Non compound word filler
hospice	Non compound word filler
refusal	Non compound word filler
polygon	Non compound word filler
modality	Non compound word filler
ganglion	Non compound word filler
nepotism	Non compound word filler
meteoric	Non compound word filler
aberrant	Non compound word filler
gluttony	Non compound word filler
asterisk	Non compound word filler
monarchy	Non compound word filler
symmetry	Non compound word filler
maternal	Non compound word filler
opposing	Non compound word filler
petition	Non compound word filler
ontology	Non compound word filler
doctrinal	Non compound word filler
amplitude	Non compound word filler
moderator	Non compound word filler
rheumatic	Non compound word filler
quotation	Non compound word filler

String of letters presented	Type of stimulus
extremist	Non compound word filler
enclosure	Non compound word filler
actuality	Non compound word filler
orchestra	Non compound word filler
secretion	Non compound word filler
intestine	Non compound word filler
odontology	Non compound word filler
convulsion	Non compound word filler
expressive	Non compound word filler
saturation	Non compound word filler
hemoglobin	Non compound word filler
psychology	Non compound word filler
oceanography	Non compound word filler
nutritionist	Non compound word filler
purification	Non compound word filler
procastinators	Non compound word filler
niesof	Nonword filler with monomorphemic-like structure
runtly	Nonword filler with monomorphemic-like structure
megatlo	Nonword filler with monomorphemic-like structure
terisma	Nonword filler with monomorphemic-like structure
trovers	Nonword filler with monomorphemic-like structure
plicits	Nonword filler with monomorphemic-like structure
assiert	Nonword filler with monomorphemic-like structure
precial	Nonword filler with monomorphemic-like structure
diamint	Nonword filler with monomorphemic-like structure
cologic	Nonword filler with monomorphemic-like structure
effinoil	Nonword filler with monomorphemic-like structure
tatchwor	Nonword filler with monomorphemic-like structure
sourcest	Nonword filler with monomorphemic-like structure
tionider	Nonword filler with monomorphemic-like structure
trophims	Nonword filler with monomorphemic-like structure

String of letters presented	Type of stimulus
tractivi	Nonword filler with monomorphemic-like structure
descenss	Nonword filler with monomorphemic-like structure
molgrass	Nonword filler with monomorphemic-like structure
tronomic	Nonword filler with monomorphemic-like structure
dosteron	Nonword filler with monomorphemic-like structure
earthfull	Nonword filler with monomorphemic-like structure
elfilmen	Nonword filler with monomorphemic-like structure
ervation	Nonword filler with monomorphemic-like structure
framplis	Nonword filler with monomorphemic-like structure
ricultrur	Nonword filler with monomorphemic-like structure
gisticoat	Nonword filler with monomorphemic-like structure
tommutals	Nonword filler with monomorphemic-like structure
texteenex	Nonword filler with monomorphemic-like structure
ervolemia	Nonword filler with monomorphemic-like structure
pressdite	Nonword filler with monomorphemic-like structure
pansition	Nonword filler with monomorphemic-like structure
vangeliss	Nonword filler with monomorphemic-like structure
milatiocy	Nonword filler with monomorphemic-like structure
phenylmon	Nonword filler with monomorphemic-like structure
letererolit	Nonword filler with monomorphemic-like structure
wrismatics	Nonword filler with monomorphemic-like structure
tantaneous	Nonword filler with monomorphemic-like structure
erfeinting	Nonword filler with monomorphemic-like structure
mertractor	Nonword filler with monomorphemic-like structure
complishly	Nonword filler with monomorphemic-like structure
ellirtatious	Nonword filler with monomorphemic-like structure
unttertracts	Nonword filler with monomorphemic-like structure
quenciality	Nonword filler with monomorphemic-like structure
quiellfortress	Nonword filler with monomorphemic-like structure
anycup	Nonword filler with compound-like structure
sinpig	Nonword filler with compound-like structure

String of letters presented	Type of stimulus
roadbud	Nonword filler with compound-like structure
clogpaw	Nonword filler with compound-like structure
postsoy	Nonword filler with compound-like structure
limebeg	Nonword filler with compound-like structure
cowfrog	Nonword filler with compound-like structure
diveran	Nonword filler with compound-like structure
vanring	Nonword filler with compound-like structure
wigmale	Nonword filler with compound-like structure
bondjude	Nonword filler with compound-like structure
porchfar	Nonword filler with compound-like structure
agonygem	Nonword filler with compound-like structure
hipopera	Nonword filler with compound-like structure
geekdawn	Nonword filler with compound-like structure
dozerjar	Nonword filler with compound-like structure
drillfox	Nonword filler with compound-like structure
ovenlimb	Nonword filler with compound-like structure
sidewolf	Nonword filler with compound-like structure
lacegolf	Nonword filler with compound-like structure
clawline	Nonword filler with compound-like structure
ponypink	Nonword filler with compound-like structure
vibewear	Nonword filler with compound-like structure
juiceloan	Nonword filler with compound-like structure
peopleraw	Nonword filler with compound-like structure
openlarva	Nonword filler with compound-like structure
beanmoral	Nonword filler with compound-like structure
gospelear	Nonword filler with compound-like structure
flirtpoor	Nonword filler with compound-like structure
jeanmedal	Nonword filler with compound-like structure
ragefiber	Nonword filler with compound-like structure
wittymain	Nonword filler with compound-like structure
tractwrap	Nonword filler with compound-like structure

String of letters presented	Type of stimulus
civilfull	Nonword filler with compound-like structure
gossipmaze	Nonword filler with compound-like structure
smileeuros	Nonword filler with compound-like structure
devourheck	Nonword filler with compound-like structure
chartsense	Nonword filler with compound-like structure
igloosuave	Nonword filler with compound-like structure
mildcounty	Nonword filler with compound-like structure
betersilica	Nonword filler with compound-like structure
womenswitchs	Nonword filler with compound-like structure
frencharctic	Nonword filler with compound-like structure
princessyawning	Nonword filler with compound-like structure

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